Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Solve 1.5x = 4.80.

2. Expand 4(x + 5).

3. Solve 7x = 56.

4. Find 3n + 1 when n = 4.

5. Find 10 − 1.286.

6. Find 0.25 × 0.04.

7. Find 0.25 × 0.02.

8. 12 shells are 30% of a collection. Find the whole collection.

9. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

10. Lena wrote 5(z + 2) = 5z + 2. Find the mistake and repair it.

11. A packet contains 9 cards. Write the number in p packets plus 4 loose cards; evaluate at p = 8.

12. Evaluate 4t + 9 when t = 8. Show the multiplication first.

13. Evaluate 4^2 + 9. Explain what the exponent means.

14. Expand 4(x + 9) and check at x = 8.

15. Simplify 9y + 4y + y. Evaluate at y = 2.

16. A rectangle has height 4 and width z + 9. Write its area two ways.

17. Solve x + 9 = 17. Show a change to both sides.

18. Solve 4x = 32. Check using multiplication.

CHALLENGE / Explain your plan and check the result

C1. A craft club chooses between A: $12 setup plus $3 per bracelet and B: $5 per bracelet. Find both costs for 4 and 8 bracelets. Find a number of bracelets for which the costs match and explain your reasoning.

C2. A student claims 2(x + 5) + 3x equals 7x for every x because 2 + 5 = 7. Test x = 2, repair the expression, and find whether the two expressions happen to agree for any positive whole-number input.